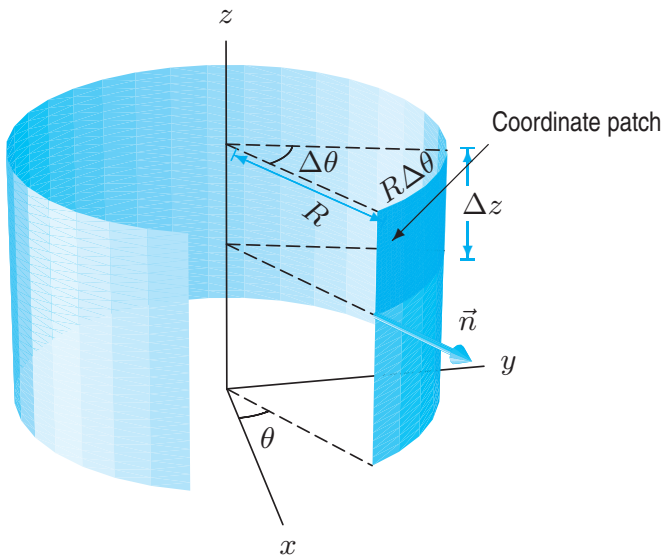
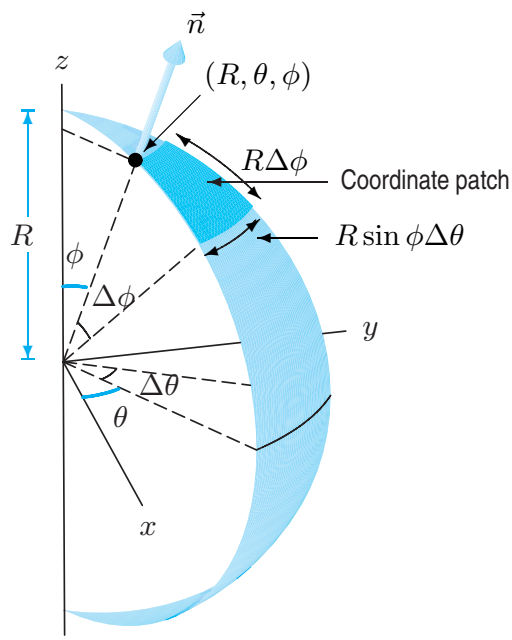
Suggested problems §19.3: 3-9 odd

Suggested problems §19.4: 1-19 odd

Problem 1: Using the diagrams below, compute the vector area element $d\vec{A}$ for cylindrical and spherical surfaces. Assume the cylinder and sphere have radius R.





Problem 2: Compute the flux of

$$\vec{F}(x, y, z) = \langle x, y, 3xz^2 \rangle$$

through the surface S of a cylinder of radius 10, centered on the z-axis, oriented outward, with $0 \leq z \leq 5$.

Problem 3: Compute the flux of

$$\vec{F}(x, y, z) = \langle 0, 0, z \rangle$$

through the sphere S of radius 5, centered at the origin, and oriented outward.

Problem 4: Compute the divergence $\nabla \cdot \vec{F}$ for the following vector fields.

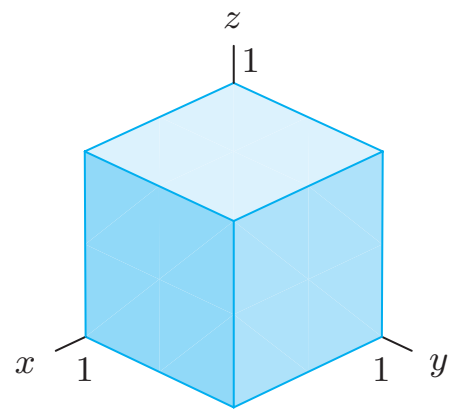
(a) $\vec{F}(x, y, z) = \langle -x + y, y + z, -z + x \rangle$

(b) $\vec{F}(x, y, z) = 3x^2\vec{i} - \sin(xz)\vec{j} + 3x^2yz^2\vec{k}$

Problem 5: Compute the flux of the vector field

$$\vec{F}(x, y, z) = (x^2 + y^2)\vec{i} + (y^2 + z^2)\vec{j} + (x^2 + z^2)\vec{k}$$

through the surface S of the cube below (oriented outward).



Problem 6: Compute the flux of $\vec{F}(x, y, z) = \langle xy, yz, xz \rangle$ through the sphere S of radius 1 centered at the origin (oriented outward).

Problem 7: Let S be an outward-oriented closed cylinder of height 2 with its base a disk of radius 2 in the xy-plane centered at the origin. Compute the flux of $\vec{F}(x, y, z) = \langle 0, y, 0 \rangle$ through S in two different ways.